PHYSIOLOGY OF

Ophthalmology | 3rd Year MBBS

Hi Sammy!
Visual pathway = most asked neuro-ophthal Q!
Diagrams are KEY!

Physiology | Vision | Ophthalmology | 3rd Year

What's Inside

- Visual Pathway & Field Defects p.2
- Accommodation & Pupillary Reflexes p.3
- Color Vision & Dark Adaptation p.4
- Visual Acuity & Errors of Refraction p.5
- Exam Strategy & High Yield p.6

Visual Pathway + Field Defects

Visual pathway = retina
to occipital cortex.
Draw this in EVERY exam!

Visual Pathway (6 Neurons/Stations)

#	Station	Key Detail
1	Photoreceptors	Rods (120M, scotopic) + Cones (6M, photopic)
2	Bipolar cells	Connect photoreceptors to ganglion cells
3	Ganglion cells	Axons form the optic nerve
4	Optic chiasma	Nasal fibers CROSS, temporal fibers UNCROSSED
5	Lateral geniculate body	Relay station in thalamus (6 layers)
6	Visual cortex (Area 17)	Calcarine fissure, occipital lobe (Brodmann 17)

Lesions + Visual Field Defects

Lesion Site	Field Defect	Cause
Optic nerve	Unilateral blindness	Optic neuritis, trauma
Optic chiasma (center)	Bitemporal hemianopia	Pituitary tumor (#1)
Optic tract	Homonymous hemianopia	Stroke, tumor
Temporal lobe (Meyer's)	Superior quadrantanopia	Pie in the sky
Parietal lobe	Inferior quadrantanopia	Pie on the floor
Occipital cortex	Homonymous hemianopia	PCA stroke, macular sparing

★ RAPD (Marcus Gunn Pupil)
Relative Afferent Pupillary Defect = optic nerve lesion. Tested by swinging flashlight test. Affected eye DILATES when light swung to it (paradoxical).

Pupillary reflexes =
asked in neuro + ophthal!
Double the marks!

Accommodation (Near Reflex Triad)

1. Convergence:	Medial recti contract (CN III)
2. Accommodation:	Ciliary muscle contracts -> lens becomes convex
3. Pupil constriction:	Sphincter pupillae contracts (parasympathetic CN III)

Pupillary Light Reflex — Pathway

Limb	Pathway	Detail
Afferent	Retina -> Optic nerve -> Chiasma	CN II carries the sensory signal
Center	Pretectal nucleus (midbrain)	Bilateral connection to EW nucleus
Efferent	EW nucleus -> CN III -> ciliary gg	Parasympathetic to sphincter pupillae

Direct reflex: Light in eye A -> constriction of pupil A
Consensual reflex: Light in eye A -> constriction of pupil B (via bilateral EW)

Important Pupillary Abnormalities

Condition	Finding	Cause
Argyll Robertson	No light reflex, intact near	Neurosyphilis (prostitute's pupil)
Marcus Gunn (RAPD)	Paradoxical dilation	Optic nerve disease
Horner syndrome	Miosis + ptosis + anhidrosis	Sympathetic chain lesion
Adie's tonic pupil	Dilated, sluggish to light	Ciliary ganglion damage
Holmes-Adie	Tonic pupil + absent DTRs	Young women, benign

★ Argyll Robertson = ARP
ARP = Accommodation Reflex Present, light reflex absent. 'Prostitute pupil' (accommodates but won't react!)

Color vision defects =
SAQ + MCQ favorite!
Know Ishihara test well!

Color Vision — Theories

Theory	Detail
Young-Helmholtz	Trichromatic — 3 cone types: Red (L), Green (M), Blue (S)
Hering	Opponent process — R-G, B-Y, Black-White pairs
Modern view	Both theories correct at different levels of processing

Color Vision Defects (Color Blindness)

Type	Defective Cone	Detail
Protanopia	Red cone absent	Red-green confusion, X-linked recessive
Deuteranopia	Green cone absent	Most common color defect, X-linked
Tritanopia	Blue cone absent	Very rare, autosomal dominant
Achromatopsia	All cones absent	Total color blindness, rod monochromatism

Tests for Color Vision

Test	Detail
Ishihara plates	Screening test — pseudoisochromatic plates (R-G defects)
Farnsworth D-15	Arrangement test — grades severity
Nagel anomaloscope	Gold standard — matches red + green to yellow
Edridge-Green lantern	Occupational test (railways, aviation)

Dark Adaptation

- Cones adapt in 7-8 min (fast), Rods adapt in 30-40 min (slow but more sensitive)
- Rhodopsin (visual purple) in rods — regenerates in dark, bleached by light
- Kohlrausch kink: point on dark adaptation curve where rod function takes over
- Vitamin A deficiency -> impaired rhodopsin regeneration -> night blindness (XN)

VA testing = clinical
posting must-know!
Snellen chart = classic!

Visual Acuity (VA) Testing

Test	Detail
Snellen chart	Standard — tested at 6m, normal = 6/6
LogMAR chart	More accurate, used in research (0.0 = 6/6)
E-chart / Landolt C	For illiterate patients
Near vision	Jaeger chart / N-notation at 33cm
For infants	Preferential looking, OKN drum, VEP

★ **6/6 = normal | 6/60 = can see at 6m what normal sees at 60m**
If VA < 6/60: test at 3m, 1m, then CF, HM, PL, No PL
CF = Counting Fingers | HM = Hand Movements | PL = Perception of Light

Rods vs Cones (Comparison)

Feature	Rods	Cones
Number	120 million	6 million
Location	Periphery (absent at fovea)	Concentrated at fovea
Function	Scotopic (dim light)	Photopic (bright light + color)
Pigment	Rhodopsin (Vit A dependent)	Iodopsin (3 types)
Sensitivity	High (1 photon can activate)	Low (need more light)
Acuity	Low (more convergence)	High (1:1 at fovea)

★ **Purkinje Shift**
In dim light, blue objects appear brighter than red (rod sensitivity peak at 505nm vs cone 555nm).

Physiology questions come
in both theory + viva!
Diagrams = extra marks!

Most Repeated LAQs (10–12 marks)

- 1. Describe the visual pathway with a diagram. Describe lesions & field defects
- 2. Describe the pupillary light reflex pathway. Add a note on RAPD
- 3. Describe the mechanism of accommodation. What is presbyopia?
- 4. Describe color vision. Classify color blindness with tests

Most Repeated SAQs (5–6 marks)

- 1. Pupillary light reflex — direct and consensual
- 2. Argyll Robertson pupil vs Horner syndrome
- 3. Dark adaptation — curve and role of Vitamin A
- 4. Rods vs Cones (table format)
- 5. Near reflex triad (accommodation reflex)
- 6. Visual acuity testing methods

One-Liner Associations (MCQ/Viva Gold)

Question	Answer
Pituitary tumor -> field defect	Bitemporal hemianopia
Most common color blindness	Deuteranopia (green cone)
Screening test for color vision	Ishihara pseudoisochromatic plates
Gold standard for color vision	Nagel anomaloscope
Accommodation reflex center	Edinger-Westphal nucleus
Pupil that accommodates but won't react	Argyll Robertson (neurosyphilis)
Miosis + ptosis + anhidrosis	Horner syndrome
Visual cortex location	Area 17, calcarine fissure, occipital
Photoreceptor for dim light	Rods (rhodopsin)
Macular sparing seen in	Occipital cortex lesion (PCA)
RAPD tested by	Swinging flashlight test

♥ **Understanding vision physiology makes you see the bigger picture, Sammy!**
Every reflex, every pathway — it all connects. You're amazing!